

JPE Replication Report

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1 Paper Identification

ID	20240482
Author	Jeenas
Title	Firm Balance Sheet Liquidity, Monetary Policy Shocks, and Investment Dynamics
Iteration	1

Some useful information up front:

- [JPE Data and Code Availability Policy](#)
- [Step by step guidance from Data Editor for Authors](#)
- [Template README](#)

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

3 General

- ☒ Data Availability and Provenance Statements
 - ☒ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☒ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☒ License for Data (optional, but recommended)
 - ☒ Details on each Data Source
- ☒ Dataset list
- ☒ Computational requirements
 - ☒ Software Requirements
 - ☒ Controlled Randomness (as necessary)
 - ☒ Memory, Runtime, and Storage Requirements
- ☒ Description of programs/code
 - ☒ License for Code (Optional, but recommended)
 - ☒ Details (as necessary)
- ☒ List of tables and programs
- ☒ References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- ☒ uses observational data.
- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
 - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.

- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README.

{REQUIRED} Please specify how long the data will be preserved in the restricted-access location.

{REQUIRED} Please provide affirmation of support for replication checks. – As per the [JPE policy](#), “In cases where data cannot be published in an openly accessible trusted data repository like the JPE dataverse, authors who have requested an exemption to publish them at the time of first submission must commit to preserving the data and code for a period of no less than five years following the publication of the paper. They should also provide reasonable assistance to requests for clarification and replication.”

{REQUIRED} Please provide a codebook for the data. – The codebook should at a minimum describe the variables obtained from the data source, in a manner that will let others verify that they have obtained a substantially similar dataset if successfully obtaining access to the data.

5.1 Data Sources

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
data/raw/FEDFUNDS_Q.xls ; data/raw/FEDFUNDS_Q.csv (Federal funds rate, FRED/Federal Reserve)	Yes	No	FRED Series FEDFUNDS	Yes (access date provided)	Yes	No
data/raw/DGS1.csv (1-year Treasury constant maturity rate, FRED/Federal Reserve)	Yes	No	FRED Series DGS1	Yes (access date provided)	Yes	No
data/raw/TB3MS.csv (3-month Treasury bill rate, FRED/Federal Reserve)	Yes	No	FRED Series TB3MS	Yes (access date provided)	Yes	No
data/raw/GDPDEF_Q.csv (GDP deflator, FRED/BEA)	Yes	No	FRED Series GDPDEF	Yes (access date provided)	Yes	No
data/raw/GDPr_Q_BEA.xls ; data/raw/GDPr_Q_BEA_Sheet0.csv ; data/raw/GDPr_Q_BEA_Sheet1.csv (BEA real GDP)	Yes	No	BEA website	Yes (access date provided)	Yes	No
data/raw/GVADEF_Q_BEA.xls ; data/raw/GVADEF_Q_BEA_Sheet0.csv ; data/raw/GVADEF_Q_BEA_Sheet1.csv (BEA GVA/value-added deflator)	Yes	No	BEA website	Yes (access date provided)	Yes	No
data/raw/LPagg_data/PCECC96.* , GPDIC1.* , GCEC1.* , DNDGRD3Q086SBEA.* , A006RD3Q086SBEA.* , A009RD3Q086SBEA.* , Y033RD3Q086SBEA.* (aggregate consumption, investment, government spending, and deflators; FRED/BEA)	Yes	No	FRED/BEA sources	Yes (access date provided)	Yes	Luetticke (2021) article, but no data citation

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
data/raw/FRB_Z1.csv (Federal Reserve Financial Accounts / Flow of Funds Z.1)	Yes	No	Federal Reserve Z.1	Yes (access date provided)	Yes	No
data/raw/bds2018.csv ; data/raw/bds2018_fage.csv (U.S. Census Bureau Business Dynamics Statistics, 2018 release)	Yes	No	Census BDS data page	Yes (access date provided)	Yes	No
data/raw/JK_data_fig4.csv (Jarocinski-Karadi monetary policy shocks)	Yes	No	replication data DOI	Yes (access date provided)	Yes	Jarocinski and Karadi (2020) article, but no data citation
data/raw/sfrr_RR.csv (Romer-Romer monetary policy shocks, robustness)	Yes	No	Ramey replication zip	Yes (access date provided)	Yes	Romer and Romer (2004) and Ramey (2016), but no data citation
proprietary-data-not-for-publication/raw/CS_data_raw.csv (Compustat North America Quarterly)	No	Yes	-	Yes (institutional access required)	Yes	No
proprietary-data-not-for-publication/raw/CRSP_data_raw.csv (CRSP U.S. Stock Database)	No	Yes	-	Yes (institutional access required)	Yes	No
proprietary-data-not-for-publication/raw/CCM_data_raw.csv (CRSP/Compustat Merged links)	No	Yes	-	Yes (institutional access required)	Yes	No
proprietary-data-not-for-publication/raw/Wscope_data_raw.xlsx (Worldscope / Refinitiv Datastream incorporation and foundation dates)	No	Yes	-	Yes (institutional access required)	Yes	No

6 Requirements

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☐ Title conforms to guidance (starts with "Data and Code for: Title of Paper" or "Code for: Title of Paper", is properly capitalized)
- ☒ Authors (with affiliations) are listed in the same order as on the paper

{REQUIRED} Please review the title of the deposit as per our guidelines.

6.1 Stated Requirements

- ☒ Software Requirements specified as follows:
 - macOS Sequoia 15.4.1 (24E263)
 - Python 3.9.5
 - R 4.1.0
 - Julia 1.9.1

- MATLAB, version 9.12.0.1956245 (R2022a) Update 2
- ☒ Computational Requirements specified as follows:
 - 3.2 GHz 6-Core Intel Core i7 processor
 - 64 GB of RAM
 - Approximately 120 GB of available storage
- ☒ Time Requirements specified as follows:
 - about 22 hours
- ☐ Requirements are complete.

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

 We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals.* If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this – please check!

Summary: - Data files with PII indicators: 5 - Variables flagged in data: 106 - Code files with PII references: 23 - PII references in code: 847

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	FRB_Z1.csv	98	lon, house, son
Data	Wscope_data_raw.xlsx	1	name
Data	bds2018.csv	3	birth, loc, location
Data	bds2018_fage.csv	3	birth, loc, location

See [Appendix](#) for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-06-08T14:40:59.206

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any [windows](#) filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that [STATA](#) allows [/](#) to be a filepath separator on a windows platform, hence this is a requirement for *all* [STATA](#) applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (C: \ etc)
48	0	13	1	0

7.1.3 Duplicate Files Report

We found the following duplicate files:

Filename	Size (MB)	Checksum (MD5)
/__MACOSX/JPE_REPLICATION_FINAL/code/._.DS_Store	0.0	0bed7e90c2bade9763fa18f1fb4441d31f91c87c
/__MACOSX/JPE_REPLICATION_FINAL/code/Julia/._.DS_Store	0.0	0bed7e90c2bade9763fa18f1fb4441d31f91c87c
/__MACOSX/JPE_REPLICATION_FINAL/code/MATLAB/._.DS_Store	0.0	0bed7e90c2bade9763fa18f1fb4441d31f91c87c

Filename	Size (MB)	Checksum (MD5)
/__MACOSX/JPE_REPLICATION_FINAL/code/Python/._.DS_Store	0.0	0bed7e90c2bade9763fa18f1fb4441d31f91c87c
/__MACOSX/JPE_REPLICATION_FINAL/code/R/._.DS_Store	0.0	0bed7e90c2bade9763fa18f1fb4441d31f91c87c
/__MACOSX/JPE_REPLICATION_FINAL/data/._.DS_Store	0.0	0bed7e90c2bade9763fa18f1fb4441d31f91c87c
/__MACOSX/JPE_REPLICATION_FINAL/data/raw/._.DS_Store	0.0	0bed7e90c2bade9763fa18f1fb4441d31f91c87c
/__MACOSX/JPE_REPLICATION_FINAL/data/raw/LPagg_data/._.DS_Store	0.0	0bed7e90c2bade9763fa18f1fb4441d31f91c87c
/__MACOSX/JPE_REPLICATION_FINAL/proprietary-data-not-for-publication/._.DS_Store	0.0	0bed7e90c2bade9763fa18f1fb4441d31f91c87c
/__MACOSX/JPE_REPLICATION_FINAL/proprietary-data-not-for-publication/raw/._.DS_Store	0.0	0bed7e90c2bade9763fa18f1fb4441d31f91c87c
/JPE_REPLICATION_FINAL/code/R/.DS_Store	0.01	df2fbeb1400acda0909a32c1cf6bf492f1121e07
/JPE_REPLICATION_FINAL/data/raw/LPagg_data/.DS_Store	0.01	df2fbeb1400acda0909a32c1cf6bf492f1121e07
/JPE_REPLICATION_FINAL/proprietary-data-not-for-publication/raw/.DS_Store	0.01	df2fbeb1400acda0909a32c1cf6bf492f1121e07
/JPE_REPLICATION_FINAL/data/raw/GVADEF_Q_BEASheet1.csv	0.02	5d55bd51e5f1718e76d07db750a10396c4d1944d

7.1.4 Zero Size Files Report

We did not find any zero sized files.

7.1.5 Large Files Report

We found the following files larger than 100MB:

Filename	Size (MB)
/JPE_REPLICATION_FINAL/proprietary-data-not-for-publication/raw/CS_data_raw.csv	529.44
/JPE_REPLICATION_FINAL/proprietary-data-not-for-publication/raw/CRSP_data_raw.csv	10212.45

7.2 Code description

The replication contains a Main Python script Master_main.py calling the main Python, Julia, R and Matlab pipelines in two sections, Empirics and Model. In addition to this, package provides language-specific scripts to install required dependencies.

cloc	github.com/AIDanial/cloc v 2.06 T=1.00 s (46.1 files/s, 1071665.9 lines/s)			
Language	files	blank	comment	code
CSV	20	0	0	1059021
Julia	5	808	1521	4093
R	4	307	535	1888
Python	13	170	305	720
MATLAB	2	36	44	202
Text	2	4	0	22

Language	files	blank	comment	code
---	---	---	---	---
SUM:	46	1325	2405	1065946

- ✓ The replication package contains a "main" or "master" file(s) which calls all other auxiliary programs.

7.3 Computing Environment of the Replicator

- Nuvolos Cloud, with standard configuration 16 cores and 64GB RAM, scaled up to 120 cores and 448GB of RAM to run model in Julia. Storage quota increased up to 150GB to accommodate model output. To ease comparisons with authors' setup, we report here also the list of dependencies used, which were installed following the README instructions.
- Python 3.9.23 using
 - pandas==1.3.1
 - numpy==1.24.4
 - scipy==1.8.1
 - openpyxl==3.1.5
 - xlrd==2.0.2
 - pytz==2026.2
 - python-dateutil==2.9.0post0
- Julia 1.9.1
 - AverageShiftedHistograms==0.8.9
 - Binscatters==0.4.0
 - CSV==0.10.14
 - CompEcon==0.4.0
 - DataFrames==1.5.0
 - Distributions==0.25.45
 - GLM==1.9.0
 - Interpolations==0.15.1
 - JLD==0.13.5
 - LaTeXStrings==1.3.1
 - Plots==1.40.8
 - QuantEcon==0.16.6
 - Roots==2.2.1
- R 4.1.2
 - plm==2.4-1
 - plyr==1.8.6
 - lmtest==0.9-38
 - stargazer==5.2.2
 - lfe==2.8-6
 - moments==0.14
 - xtable==1.8.4
 - fixest==0.10.1
- Matlab R2022a

8 Replication

- Downloaded code and data from dropbox and cloned Git repo.
- Instantiated Julia, Python and R dependencies following Readme. Our cloud machine makes it complex to run all within Python, as cross-application interaction is limited. Therefore, replicator proceeded running master script tasks separately by language, following the script's order.
- Adjustment before running run_python(PROJECT_ROOT / "code" / "Python" / "FOFA_analysis_code.py") for missing dependencies pytz and dateutil. Installed via conda install pytz and conda install python-dateutil. Ran again. Authors might want to add these two package dependencies to their script and README.

- Running “JPE_REPLICATION_FINAL/code/Julia/Julia_model_code/Master_model_work.jl” required more resources than described by README. We ran out of RAM memory with 64 and 128 GB on a Linux-based system. Also, storage needed to be increased above 120 GB.
- Ran the rest of the code successfully.

9 Findings

9.1 Missing Requirements

- ☒ Software Requirements

☐ Python packages

 - pytz
 - dateutil

[REQUIRED] Please amend README to contain complete requirements.

Additionally, authors are invited to doublecheck RAM and storage requirements, as running “JPE_REPLICATION_FINAL/code/Julia/Julia_model_code/Master_model_work.jl” required more resources than described by README. We ran out of RAM memory with 64 and 128 GB on a Linux-based system. Also, allocated storage needed to be increased to 150 GB.

9.2 Data Preparation Code

Data Preparation Code ran without errors.

9.3 Tables and Figures

Replication successfully generated all tables and figures in the paper. All replicated output matched latest draft of the paper.

Figure/Table #	Program	Line Number	Reproduced
Figure 1 (Fig1a, Fig1b)	code/R/Micro_reg_code.r	404, 415, 425	Yes
Figure 2 (Fig2a-Fig2c)	code/R/Micro_eventstudy_plottr.r	68	Yes
Figure 3 (Fig3a, Fig3b)	code/R/Micro_reg_code.r	1387	Yes
Figure 4 (Fig4a, Fig4b)	code/Julia/Julia_model_code/Master_model_work.jl	280, 431	Yes
Figure 5	code/Julia/Julia_model_code/Master_model_work.jl	579	Yes
Figure 6	code/Julia/Julia_model_code/auxiliary_model_functions.jl	2487	Yes
Figure 7 (Fig7a, Fig7b)	code/Julia/Julia_model_code/Master_model_work.jl	1104, 1127	Yes
Figure 8 (Fig8a, Fig8b)	code/Julia/Julia_model_code/Master_model_work.jl	1161, 1171	Yes
Table 1	code/R/Micro_reg_code.r	746	Yes
Table 2	code/Julia/Julia_model_code/Master_model_work.jl	831	Yes
Figure A.1	code/R/Micro_reg_code.r	404	Yes
Table A.1	code/R/Micro_reg_code.r	1005	Yes
Table A.2	code/R/Micro_reg_code.r	517	Yes
Table A.3	code/R/Micro_reg_code.r	612	Yes

Figure/Table #	Program	Line Number	Reproduced
Table A.4	code/R/Micro_reg_code.r	679	Yes
Figure A.2 (FigA2a- FigA2c)	code/R/Micro_eventstudy_plottr.r	98	Yes
Figure A.3 (FigA3a- FigA3c)	code/R/Micro_eventstudy_plottr.r	98	Yes
Figure A.4 (FigA4a- FigA4c)	code/R/Micro_eventstudy_plottr.r	98	Yes
Figure A.5 (FigA5a, FigA5b)	code/R/Micro_reg_code.r	1387, 1922	Yes
Figure A.6 (FigA6a- FigA6d)	code/R/Micro_reg_code.r	1387, 1922	Yes
Figure A.7 (FigA7a, FigA7b)	code/R/Micro_reg_code.r	1387	Yes
Figure A.8 (FigA8a, FigA8b)	code/R/Micro_reg_code.r	1387	Yes
Figure A.9 (FigA9a, FigA9b)	code/R/Micro_reg_code.r	1387	Yes
Figure A.10 (FigA10a, FigA10b)	code/R/Micro_reg_code.r	1405	Yes
Figure A.11 (FigA11a, FigA11b)	code/R/Micro_reg_code.r	1387	Yes
Figure A.12 (FigA12a, FigA12b)	code/R/Micro_reg_code.r	1387, 1756	Yes
Figure A.13 (FigA13a, FigA13b)	code/R/Micro_reg_code.r	1756	Yes
Figure A.14 (FigA14a, FigA14b)	code/R/Micro_reg_code.r	2212, 2237	Yes
Table A.5	code/Python/FOFA_analysis_code.py	107	Yes
Figure A.15 (FigA15a- FigA15d)	code/MATLAB/LPAgg_main.m	183, 205, 227, 250	Yes
Table B.1 (manual table from components)	code/Julia/Julia_empirics_code/BDS_analysis_code.jl; code/R/Micro_reg_code.r; code/Julia/Julia_model_code/Master_model_work.jl	40; 218; 204	Yes
Table B.2	code/Python/Generate_TabB2.py	15, 60	Yes
Table B.3	code/R/Micro_model_reg_code.r	205	Yes
Table B.4	code/R/Micro_model_reg_code.r	273	Yes
Table B.5 (manual table from component)	code/Julia/Julia_model_code/calib_functions.jl	392	Yes
Figure B.1	code/Julia/Julia_model_code/calib_functions.jl	338	Yes
Figure B.2	code/Julia/Julia_model_code/auxiliary_model_functions.jl	2514	Yes

Figure/Table #	Program	Line Number	Reproduced
Figure B.3 (FigB3a, FigB3b)	code/Julia/Julia_model_code/auxiliary_model_functions.jl	2585, 2648	Yes
Figure B.4	code/Julia/Julia_model_code/auxiliary_model_functions.jl	2526	Yes
Figure B.5	code/Julia/Julia_model_code/Master_model_work.jl	1088	Yes
Figure B.6	code/Julia/Julia_model_code/Master_model_work.jl	690	Yes
Figure B.7	code/Julia/Julia_model_code/Master_model_work.jl	738	Yes
Figure B.8	code/Julia/Julia_model_code/Master_model_work.jl	1146	Yes
Figure B.9	code/Julia/Julia_model_code/Master_model_work.jl	1330	Yes
Figure B.10	code/Julia/Julia_model_code/Master_model_work.jl	612	Yes
Figure B.11 (FigB11a, FigB11b)	code/Julia/Julia_model_code/auxiliary_model_functions.jl; code/Julia/Julia_model_code/Master_model_work.jl	2492; 1245	Yes
Figure B.12	code/Julia/Julia_model_code/Master_model_work.jl	1457	Yes
Figure B.13 (FigB13a, FigB13b)	code/Julia/Julia_model_code/auxiliary_model_functions.jl; code/Julia/Julia_model_code/Master_model_work.jl	2495; 1574	Yes
Figure B.14 (FigB14a, FigB14b)	code/Julia/Julia_model_code/Master_model_work.jl	1474, 1590	Yes

9.4 In-Text Numbers

- ☐ In-text numbers not verified.
- ☐ There are no in-text numbers, or all in-text numbers stem from tables and figures.
- ☐ There are in-text numbers, but they are not identified in the code
- ☒ There are in-text numbers, and the identified in-text numbers were verified against the auxiliary output files.

All in-text numbers detected were reproduced by the replication package. Replicated in-text numbers match the ones reported in the paper.

In-text numbers	Program	Line Number	Reproduced
Footnote 3	code/R/Micro_reg_code.r	258, 313	Yes
Footnote 15	code/R/Micro_reg_code.r	308	Yes
Footnote 46	code/Julia/Julia_model_code/Master_model_work.jl	215	Yes
Section 4.2 text	code/Julia/Julia_model_code/Master_model_work.jl	288	Yes
Section 5.1 text	code/Julia/Julia_model_code/Master_model_work.jl	671	Yes
Section 5.2 text	code/Julia/Julia_model_code/Master_model_work.jl	1188, 1212, 1332	Yes
Footnote 57	code/Julia/Julia_model_code/Master_model_work.jl	1220	Yes
Appendix A.2 text	code/R/Micro_reg_code.r	520	Yes
Footnote 61	code/R/Micro_reg_code.r	791	Yes
Footnote 74	code/R/Micro_reg_code.r	327	Yes
Appendix B.5.3 text	code/Julia/Julia_model_code/Master_model_work.jl	372	Yes

In-text numbers	Program	Line Number	Reproduced
Footnote 87	code/Julia/Julia_model_code/calib_functions.jl	345	Yes
Appendix B.10.1 text	code/Python/FOFA_analysis_code.py; code/Python/TB3_analysis_code.py	111; 29	Yes
Appendix B.11.1 text	code/Julia/Julia_model_code/Master_model_work.jl	495	Yes
Appendix B.11.2 text	code/Julia/Julia_model_code/Master_model_work.jl	1616	Yes

10 Classification

- ☒ full reproduction
- ☐ full reproduction with minor issues
- ☐ partial reproduction (see above)
- ☐ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☒ None.
- ☐ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☐ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check
 - ☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☐ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements